

Advay Sinha

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EDUCATION

Bennett University

B.Tech, Computer Science and Engineering (AI Specialization) — CGPA: **8.7 / 10**

DPS Ranchi

10+2, CBSE Board — Percentage: **93%**

Greater Noida, UP, India

Aug 2024 – May 2028

Ranchi, Jharkhand, India

2022 – 2024

EXPERIENCE

Airports Authority of India — Jay Prakash Narayan International Airport

Software Engineering Intern (Networks & Cybersecurity)

- Architected and deployed **NIMS**, an AI-driven platform unifying intrusion detection, health prediction, and configuration auditing across the airport's multi-vendor campus network (Huawei, Cisco, Hirschmann, Sophos).
- Field-validated a read-only ingestion layer (SNMPv3 polling, traps, syslog, SSH, SIEM API) against a **production Huawei S5720** switch, surfacing **87 configuration findings** with rollback-verified dry-run remediation. Enforced safety by construction: no code path can modify a device.

Alignerr (Freelance)

AI Data & Evaluation Contractor

- Evaluated and labelled large-scale model-output datasets for frontier AI systems (clients including **Anthropic**), applying structured rubrics across **10+ evaluation cycles** to improve response-quality and safety benchmarks.
- Authored adversarial test cases targeting logical and algorithmic edge cases to probe model robustness across code generation, reasoning, and multi-step instruction-following tasks.

Zarva

Backend Software Developer

- Engineered a real-time speech-recognition pipeline and **5+ REST APIs** over WebSockets for live call connectivity. Enabled instant distress detection across **100+ concurrent** cab rides, reducing safety-response latency.
- Devised a graph-theoretic shortest-path routing algorithm over integrated crime datasets, cutting user risk-exposure scores by **40%** in high-incident zones.

PROJECTS

NIMS – Network Intrusion Monitoring System | Python, PyTorch, XGBoost, FastAPI, React, SHAP | GitHub

- Benchmarked **7 ML/DL models** over **44 experiments** on 3 IDS datasets (NSL-KDD, UNSW-NB15, CIC-IDS2017) through a leakage-free, seeded pipeline. Promoted XGBoost at test F1 **0.9925 / 0.9244 / 0.9988** with SHAP attribution.
- Root-caused LightGBM multiclass divergence to unregularised leaf outputs, raising F1 from **0.775 to 0.9923** (NSL-KDD) and **0.742 to 0.9987** (CIC-IDS2017). Shipped an Optuna tuning loop, model registry, FastAPI inference service, and deterministic correlation engine over **74 offline test modules**.

Algo Trade Simulator | FastAPI, React, PostgreSQL, Docker, OpenAI API | GitHub

- Developed a full-stack trading-simulation platform handling **50+ concurrent sessions**. Optimised an SMA-crossover backtesting engine over 5 years of time-series data, improving throughput **3×** at **80%+** accuracy.
- Implemented **10+ JWT-secured REST endpoints** (Swagger-documented) plus an LLM-powered research copilot scoring historical return distributions, cutting research time **60%**.

FloatChat – Oceanographic RAG Explorer | Python, Streamlit, ChromaDB, Ollama, Plotly | GitHub

- Engineered a RAG pipeline with ChromaDB cosine-similarity indexing over **10,000+ records**, cutting query latency **70%**. Added spatial/temporal querying with local-LLM fallback (Ollama) and Plotly/Cesium geospatial views.

ACHIEVEMENTS

- Smart India Hackathon 2025**: selected among **450+ competing teams** to advance to the pre-national selection stage.
- 1st Place, MOBILON App Showcase (Mobile Next)**: won among **40+ teams** for the real-time cab-safety system delivered at Zarva (WebSocket call streaming, live distress detection, risk-aware routing).

TECHNICAL SKILLS

Languages: Java, C++, Python, JavaScript, TypeScript, SQL

Core CS: Data Structures, Algorithms, OOP, System Design, Concurrency, Distributed Systems

Backend & Systems: REST APIs, WebSockets, FastAPI, Node.js, Spring Boot, JWT Auth, CI/CD, SNMP & Syslog Telemetry

Frontend & Databases: React, Next.js, HTML/CSS, PostgreSQL, MySQL, MongoDB, ChromaDB

AI / ML: PyTorch, Scikit-learn, XGBoost, LightGBM, Optuna, SHAP, RAG Pipelines, LangChain, Pandas, NumPy

Tools: Git, Linux, Docker, Swagger, Jupyter